

SALUS Calculus & Statistical Specification (Pramana Engine v1)

SALUS — *The evidence behind every path to healing.*

Purpose: This document defines the exact mathematical models, defaults, constraints, validation gates, and source-citation rules required to make SALUS mathematically rigorous, clinically interpretable, and auditable.

Status: Implementation-ready specification

1. Non-Negotiable Integrity Rules

- No numeric output without uncertainty.
- No recommendation without source-linked evidence records.
- No source without verifiable digital identifiers (PMID , DOI , NCT , CTRI , ICTRP , AYUSH , DHARA).
- No claim can be marked “high confidence” unless calibration and coverage targets pass.
- If confidence gate fails, output must be INSUFFICIENT_EVIDENCE .

2. Mathematical Core

2.1 Study-Level Contribution

For each study i , contribution to intervention-condition credibility at time t is:

$$S_i(t) = w_{\text{grade},i} \cdot w_{\text{design},i} \cdot w_{\text{bias},i} \cdot w_{\text{recency},i}(t) \cdot f(n_i) \cdot p_i$$

Where: - w_{grade} : prior trust by evidence grade. - w_{design} : method class (RCT, cohort, etc.). - w_{bias} : risk-of-bias multiplier. - $w_{\text{recency}}(t)$: time decay. - $f(n)$: sample size transform. - p_i : precision/reliability term from variance/CI.

2.2 Grade and Design Weights (Default v1)

Category	Weight
Grade A	1.00
Grade B	0.60
Grade C	0.25

Design type	Weight
Systematic review/meta-analysis of RCTs	1.00
Multi-center randomized trial	0.90
Single-site randomized trial	0.85
Cohort/case-control	0.70
Open-label/interventional non-randomized	0.60
Mechanistic/preclinical	0.35
Historic textual consensus only	0.20

2.3 Bias Multiplier (Default v1)

Risk-of-bias assessment	Multiplier
Low	1.00
Some concerns	0.80
High	0.55
Critical / unassessed	0.40

2.4 Recency Decay (Continuous Time)

$$w_{\text{recency}}(t) = e^{-\lambda \Delta t}, \quad \lambda = \frac{\ln 2}{T_{1/2}}$$

- Δt : years since publication or last verification.
- $T_{1/2}$: evidence half-life by domain.

Default $T_{1/2}$ values: - Fast-moving pharmacotherapy: 6 years - Chronic disease lifestyle interventions: 10 years - Stable physiology/mechanism literature: 12 years

2.5 Sample Size Transform

$$f(n) = \log(1 + n)$$

Rationale: monotone increasing, diminishing returns, robust to extreme n .

2.6 Precision Term

For each study estimate $\widehat{\theta}_i$ with standard error SE_i :

$$p_i = \frac{1}{SE_i^2 + \epsilon}$$

with small stabilizer $\epsilon = 10^{-6}$.

If only confidence interval is available:

$$SE_i \approx \frac{CI_{upper} - CI_{lower}}{2 \cdot 1.96}$$

2.7 Pramana Raw and 0-100 Score

$$\text{PramanaRaw}(t) = \sum_{i=1}^N S_i(t)$$

Robust normalization (winsorized at p2/p98):

$$\text{PramanaScore}(t) = 100 \cdot \frac{\text{clip}(\text{PramanaRaw}, q_{0.02}, q_{0.98}) - q_{0.02}}{q_{0.98} - q_{0.02}}$$

3. Meta-Analysis Backbone

3.1 Fixed-Effect Inverse-Variance Pooling

$$\widehat{\theta}_{FE} = \frac{\sum_i w_i \widehat{\theta}_i}{\sum_i w_i}, \quad w_i = \frac{1}{SE_i^2}$$

3.2 Random-Effects Pooling

$$\widehat{\theta}_{RE} = \frac{\sum_i w_i^* \widehat{\theta}_i}{\sum_i w_i^*}, \quad w_i^* = \frac{1}{SE_i^2 + \tau^2}$$

Where τ^2 is between-study variance (REML preferred).

3.3 Heterogeneity Metrics

- Cochran’s Q for heterogeneity test.
- I^2 for inconsistency proportion.
- Prediction interval required when $k \geq 5$ studies:

$$\widehat{\theta}_{RE} \pm t_{k-1, 0.975} \sqrt{\tau^2 + SE(\widehat{\theta}_{RE})^2}$$

4. Bayesian Confidence Gate

For clinically meaningful threshold θ_{min} :

$$P(\theta > \theta_{min} \mid D)$$

Decision gate: - **RECOMMEND** if $P(\theta > \theta_{min} \mid D) \geq \tau_{recommend}$ - **CONDITIONAL** if in gray zone - **INSUFFICIENT_EVIDENCE** otherwise

Default thresholds: - Clinician mode: $\tau_{recommend} = 0.85$ - Public mode: $\tau_{recommend} = 0.90$ - Gray zone lower bound: 0.65

5. Dose-Response Utility Optimization

Let dose be d :

$$U(d) = B(d) - \alpha R(d) - \beta C(d)$$

- $B(d)$: benefit model (e.g., Emax/Hill)
- $R(d)$: adverse-risk model
- $C(d)$: adherence/cost burden

Optimal dose solves:

$$\frac{dU}{dd} = 0, \quad \frac{d^2U}{dd^2} < 0$$

With hard constraints: - regulatory max dose - known toxicity boundaries - contraindicated co-medication combinations

If constrained optimum infeasible, choose best feasible boundary point.

6. Interaction Dynamics (ODE Safety Layer)

For two co-administered interventions:

$$\frac{dC_1}{dt} = -k_1 C_1 - \beta C_1 C_2, \quad \frac{dC_2}{dt} = -k_2 C_2$$

Risk score over time:

$$\text{Risk}(t) = h(C_1(t), C_2(t), x_{\text{patient}})$$

Output required in UI: - peak risk window - time-to-peak - interaction severity class - evidence confidence for the interaction edge

7. Personalized Trajectory Model

State vector (e.g., HbA1c, SBP, CRP):

$$\frac{dx}{dt} = F(x, t) + G(x, t)u(t) + \epsilon(t)$$

Counterfactual policy simulation required: 1. allopathy-only 2. integrative care 3. lifestyle-first

Outputs: - expected trajectory - 50% and 95% uncertainty bands - probability of crossing clinical targets by horizon T

8. Calibration and Efficacy Demonstration Protocol

8.1 Required Offline Validation

1. Discrimination

- AUROC for binary outcomes where applicable
- Concordance index for time-to-event endpoints

2. Calibration

- Calibration slope in [0.9, 1.1]
- Calibration intercept in [-0.05, 0.05]
- Brier score tracked by condition

3. Interval reliability

- 95% intervals empirical coverage: 92%-98%

4. Ranking utility

- Spearman correlation between Pramana ranking and independent review conclusions

5. Decision usefulness

- Net benefit via Decision Curve Analysis over clinician-relevant threshold range

8.2 Required Online Monitoring

- Drift detection on score distribution (PSI/KL divergence)
- Monthly re-calibration checks
- Alert when confidence gate pass rate changes by >10% week-over-week

8.3 Minimum Efficacy Criteria for Production Rollout

All must pass for two consecutive validation windows: - AUROC ≥ 0.72 (or task-appropriate equivalent) - Calibration slope in [0.9,1.1] - 95% interval coverage $\geq 92\%$ - No critical safety rule violations in dose/interaction engines

9. Honesty-by-Construction Source Policy

9.1 Citation Requirements

Every intervention card must show: - source identifiers (PMID / DOI /registry id) - direct URL - last verification date - model version used to compute score

9.2 Prohibited Outputs

System must reject generation of: - unsupported causal claims - equivalency claims across grades (A vs B/C) - recommendations with missing uncertainty - references without resolvable identifiers

9.3 Allowlisted Source Domains

- pubmed.ncbi.nlm.nih.gov
- doi.org
- clinicaltrials.gov
- trialsearch.who.int
- ctri.nic.in
- ayush.gov.in
- cochrane.org
- cochranelibrary.com
- bmj.com / bmjopen.bmj.com

10. Implementation Mapping

10.1 Data Model Additions (`packages/domain/src/index.ts`)

Add fields to `treatmentEvidenceSchema` :- `publicationDate`: string (ISO date) - `riskOfBias`: "low" | "some_concerns" | "high" | "critical" - `effectEstimate?`: number - `effectStdError?`: number - `confidenceIntervalLower?`: number - `confidenceIntervalUpper?`: number - `pramanaContribution?`: number - `pramanaScore?`: number - `modelVersion?`: string

10.2 API Endpoints (`services/api/src/app.ts`)

Add: - `GET /score/:conditionId/:intervention` -> score + uncertainty + citation list - `GET /trajectory/:conditionId` -> scenario trajectories + intervals - `GET /interaction-risk` -> ODE-based interaction timeline

10.3 Compute Jobs (`services/ingestion` or `services/scoring`)

Nightly scheduled jobs: - recalculate recency factors - recompute score normalization quantiles - refresh calibration diagnostics - emit audit artifact JSON to S3

11. Governance & Reproducibility

- Every score response must include `modelVersion` and `dataSnapshotId` .
- Every production model change requires:
 1. version bump,
 2. changelog entry,
 3. backtest report,
 4. sign-off by clinical and statistical reviewers.

12. Source References (Real, Public, Citable)

1. Oxford CEBM Levels of Evidence (official):
 - <https://www.cebm.ox.ac.uk/resources/levels-of-evidence/ocebmllevels-of-evidence>
2. Cochrane Handbook v6.5, Chapter 10 (meta-analysis, heterogeneity, Bayesian methods):
 - <https://www.cochrane.org/authors/handbooks-and-manuals/handbook/current/chapter-10>
3. NCBI Entrez E-utilities Help (official API/programmatic interface):
 - <https://www.ncbi.nlm.nih.gov/books/NBK25501/>
4. ClinicalTrials.gov Data API (official):
 - <https://clinicaltrials.gov/data-api/about-api>
5. WHO ICTRP Trial Search Portal (official):

- <https://trialsearch.who.int/>

Additional cited statistical classics: – DerSimonian R, Laird N. Controlled Clinical Trials (1986). – Higgins JPT, Thompson SG. Stat Med (2002) on I^2 . – Riley RD, Higgins JPT, Deeks JJ. BMJ (2011) on random-effects interpretation.

13. Release Milestones for Math Stack

Milestone M1 (2 weeks)

- Recency decay + score normalization + uncertainty in API response.
- Validation: calibration and coverage baseline.

Milestone M2 (4 weeks)

- Bayesian confidence gate + dose-response optimizer.
- Validation: decision-curve net benefit and safety constraints.

Milestone M3 (6 weeks)

- ODE interaction timeline + trajectory simulator.
 - Validation: retrospective agreement with known interaction timing and outcome trends.
-

14. Final Principle

If the system cannot show the equation, uncertainty, and source, it must not show the claim.